MATH3131 Time Series Analysis

Exploratory Data Analysis Report

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1 Data Description and Cleaning Process

We base this Exploratory Data Analysis (EDA) on official demographic data from Singapore, specifically looking at two key variables: Total Live Births (TLB) and the Total Fertility Rate (TFR) recorded between 1960 and 2024.

Before fitting any models, I ran a brief data cleaning process in R to ensure everything is fully reproducible and properly formatted:

- **Sorting Chronologically:** I extracted the relevant TFR and TLB rows from the raw file. Because the original data was listed backwards (2024 down to 1960), I used the `rev()` function to flip the values into a standard forward timeline.
- **Formatting as Numerics:** The data rows were then converted into simple one-dimensional vectors via `unlist()` and explicitly forced into a numeric format to prevent type errors.

Finally, I converted the clean vectors into standard `ts` objects. The data was split into a main training set (1960–2012) for modeling, while keeping the recent years (2013–2024) hidden as a test set to evaluate forecasting accuracy later.

As a minor limitation of this specific EDA scope, while the raw dataset actually provides rich, detailed breakdowns by age brackets and ethnic groups (e.g., Chinese, Malays, Indians), I deliberately extracted only the national aggregates (Total Live Births and Total Fertility Rate). By focusing exclusively on these macro-level variables, our subsequent time series models will inevitably mask any divergent demographic trends hidden within these specific sub-groups, which constrains deeper causal analysis.

2 Visualisations and Temporal Features

To understand the underlying patterns of Singapore's demographic changes, line plots were generated for both TLB and TFR covering the period from 1960 to 2024.

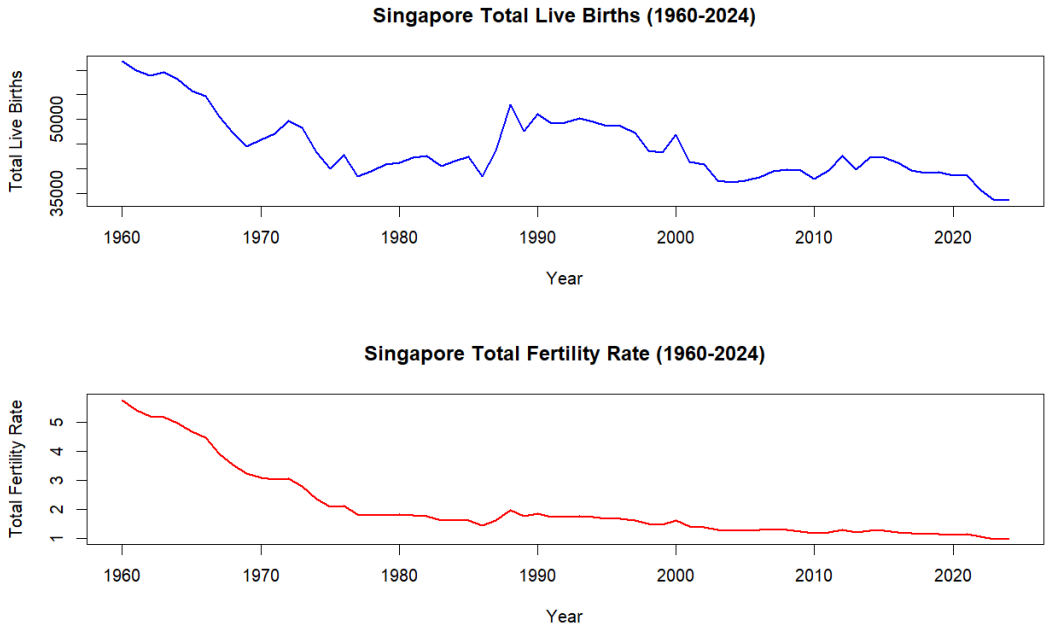

Figure 1: Raw Time Series of Singapore TLB and TFR (1960-2024)

As illustrated in Figure 1, both TLB and TFR share a distinctive, long-term downward trajectory. The TFR exhibits a sharp decline from over 5.0 in the 1960s to well below the demographic replacement level. This non-stationary behavior, characterized by a clear trend and a shifting mean, visually confirms that the raw data requires first-order differencing ($d = 1$) before stationary modeling can be applied.

3 Analysis of Time Series Features and Model Selection

3.1 Total Fertility Rate (TFR)

The temporal features of the TFR training set (1960-2012) were first evaluated. The autocorrelation function (ACF) of the original series decays at an extremely slow rate (Figure 2), confirming non-stationarity.

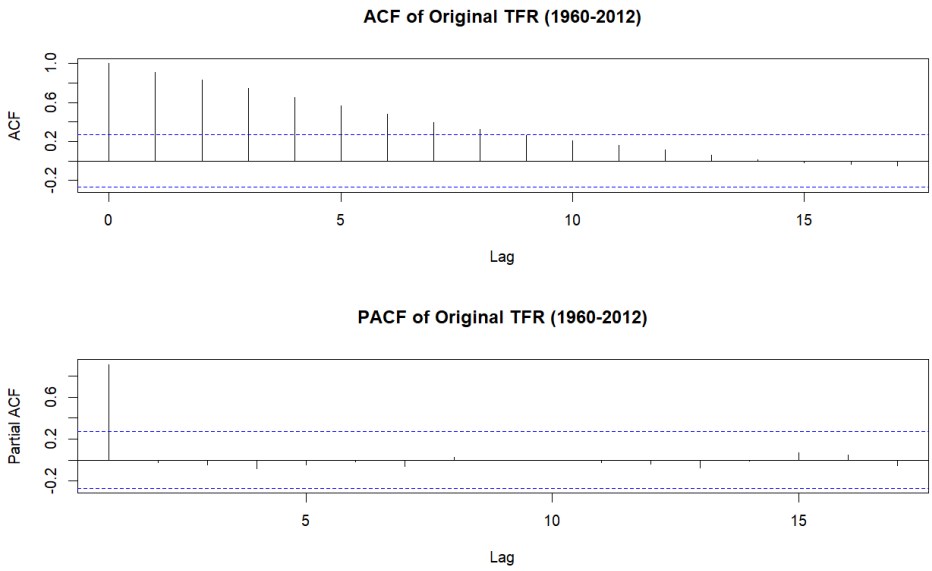

Figure 2: ACF and PACF of the Original TFR Training Set

After applying a first-order difference ($d = 1$), the stationary series was re-examined.

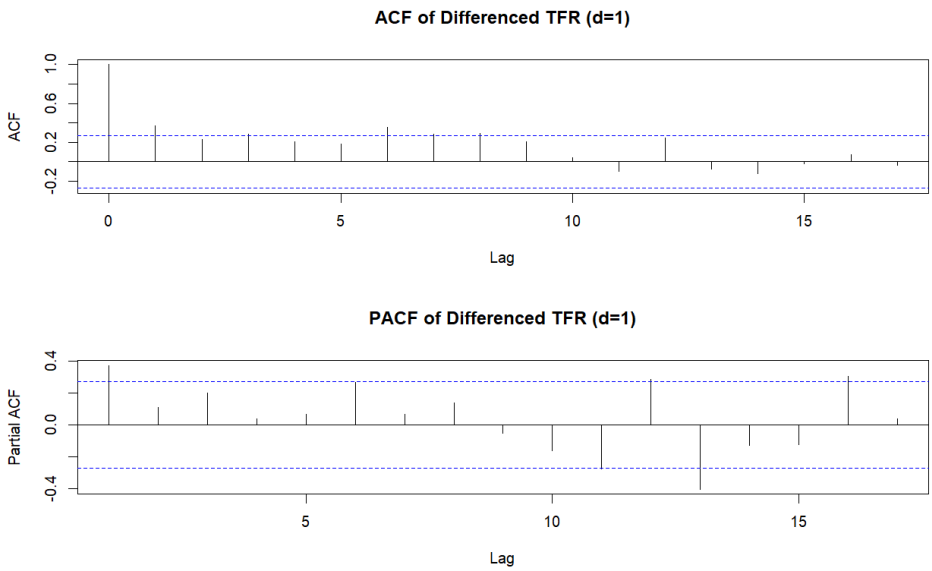

Figure 3: ACF and PACF of the Differenced TFR ($d = 1$)

As shown in Figure 3, both the ACF and PACF plots exhibit a significant spike at lag 1 before falling within the 95% confidence bounds. The significant spike at lag 1 in the PACF suggests a first-order autoregressive component ($p = 1$), while the similar spike in the ACF suggests a first-order moving average component ($q = 1$). While a minor spike appears at lag 13, it is treated as a statistical artifact (Type I error) to prevent overfitting on a small training set ($N = 53$). Based on the principle of parsimony, an ARIMA(1, 1, 1) model is proposed as the preliminary candidate for TFR.

3.2 Total Live Births (TLB) and Cultural Seasonality

Following the same differencing procedure ($d = 1$), the differenced TLB series was examined.

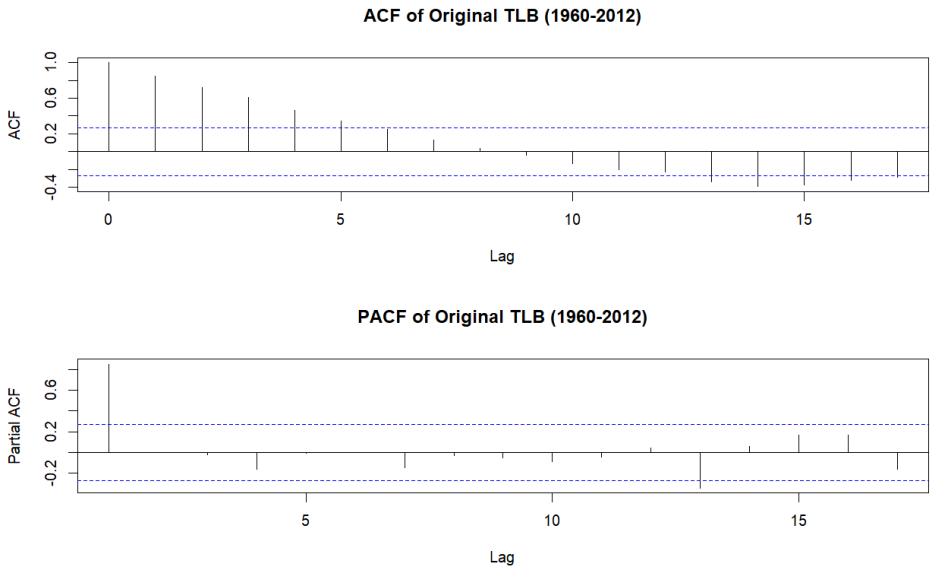

Figure 4: ACF and PACF of the Original TLB Training Set

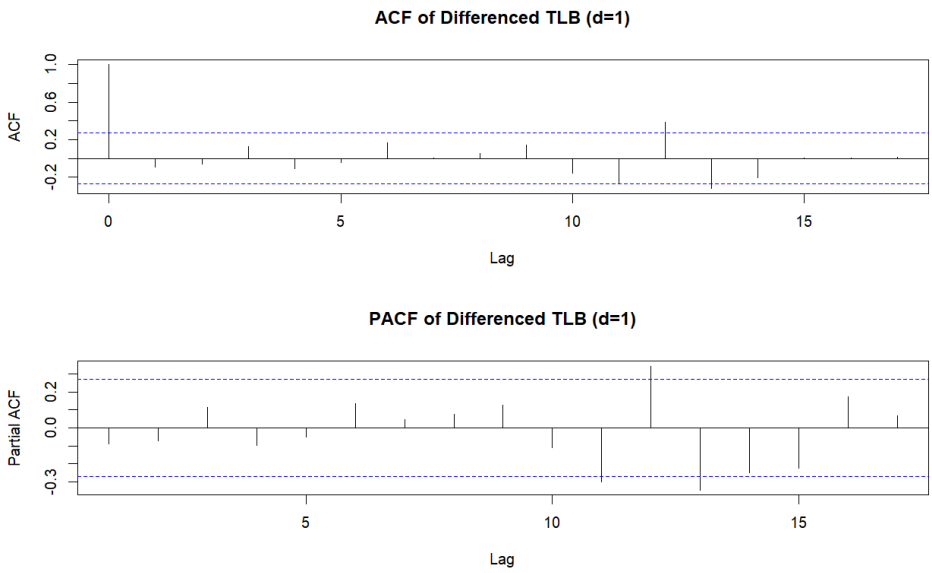

Figure 5: ACF and PACF of the Differenced TLB ($d = 1$)

Checking Figure 5, I found that the short-term autocorrelations (lags 1 to 5) are mostly insignificant. Because of this lack of immediate short-term structure, I decided to use a simple random walk, ARIMA(0, 1, 0), as my initial baseline. The most interesting feature, however, is a highly significant spike popping up exactly at lag 12 in both the ACF and PACF plots. When connecting this statistical pattern back to the real-world demographic context, this 12-year cycle perfectly reflects the cultural "Dragon Year effect"

in Singapore, where birth rates reliably experience periodic surges. To handle this accurately in the final forecasting report, I will definitely need to upgrade this baseline to a more advanced seasonal model that directly incorporates this 12-year cultural periodicity.

4 Model Diagnostics and Assessment

4.1 Residual Diagnostics & Statistical Tests

After fitting the preliminary models, residual diagnostics were performed to verify adequacy.

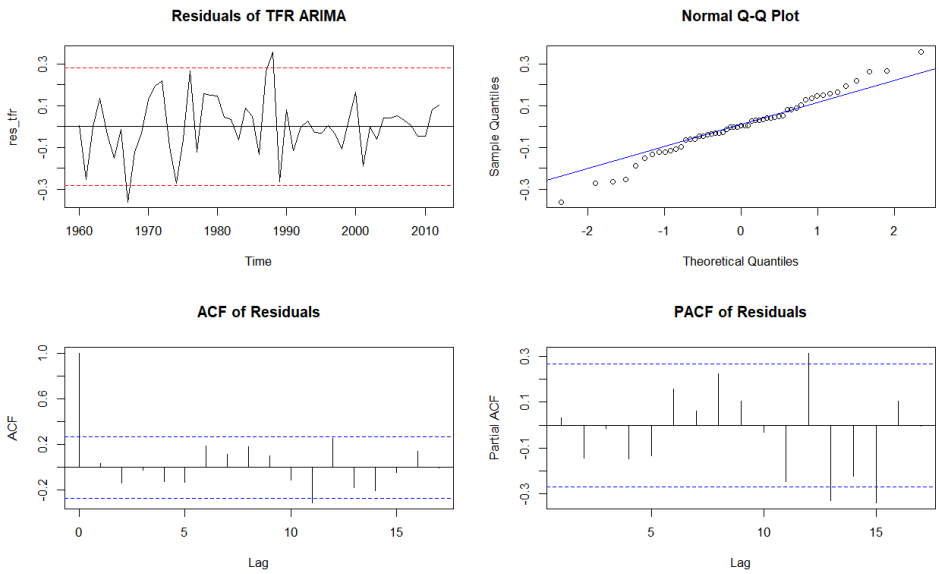

Figure 6: Residual Diagnostics for the TFR ARIMA(1,1,1) Model

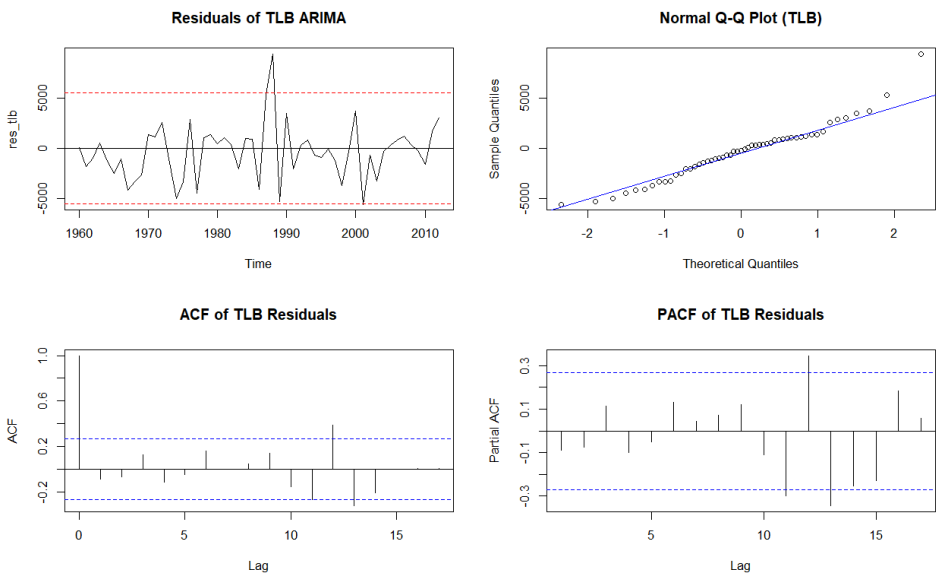

Figure 7: Residual Diagnostics for the TLB ARIMA(0,1,0) Model

Visually (Figures 6 and 7), the residuals for both models resemble white noise. The time series plots wander randomly around zero, Q-Q plots suggest rough normality, and remaining short-term lags fall within confidence intervals.

To quantitatively confirm this, a Portmanteau test (specifically, the Ljung-Box test) was conducted. The test yielded a p-value of 0.4518 for the TFR model and 0.7024 for the TLB model. Since both values are significantly greater than the 0.05 threshold, we fail to reject the null hypothesis, mathematically proving that the residuals are statistically indistinguishable from white noise.

```
> print("--- Portmanteau Test (Ljung-Box) for TFR ARIMA(1,1,1) Residuals ---")
[1] "--- Portmanteau Test (Ljung-Box) for TFR ARIMA(1,1,1) Residuals ---"
> Box.test(res_tfr, lag = 10, type = "Ljung-Box")

        Box-Ljung test

data:  res_tfr
X-squared = 9.8719, df = 10, p-value = 0.4518

>
> print("--- Portmanteau Test (Ljung-Box) for TLB ARIMA(0,1,0) Residuals ---")
[1] "--- Portmanteau Test (Ljung-Box) for TLB ARIMA(0,1,0) Residuals ---"
> Box.test(res_tlb, lag = 10, type = "Ljung-Box")

        Box-Ljung test

data:  res_tlb
X-squared = 7.2418, df = 10, p-value = 0.7024
```

Figure 8: Ljung-Box Test Results from R Console

4.2 Model Assessment Strategy for Future Forecasting

To determine which viable models are best for forecasting the period from 2013 to 2024 in the Final Report, a comprehensive evaluation strategy will be employed. Firstly, in-sample goodness-of-fit will be assessed using the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC), which strictly penalise model complexity. Secondly, out-of-sample forecasting performance will be evaluated against the test set (2013-2024). Accuracy metrics such as the Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE) will be calculated to select models that minimise prediction errors.

5 Proposed Research Questions

Based on the EDA, the primary research questions are defined as:

1. How can we effectively model the non-stationary, long-term declining trend of Singapore's Total Fertility Rate (TFR) using ARIMA-based methods to generate reliable short-term forecasts?

2. How does the 12-year cultural "Dragon Year" cycle impact the Total Live Births (TLB), and what time series modelling strategies are required to adequately capture this long-term periodicity compared to standard baseline models?

Anticipated Limitations and Difficulties:
Although the basic ARIMA framework gives us a great starting point, real-world demographic data is incredibly sensitive to unpredictable external shocks. Things like sudden

government pro-family policies, financial crashes, or COVID-19 create massive non-linear shifts. Because our current models are strictly linear, they will likely struggle to predict these sudden structural breaks. On top of that, mathematically capturing the 12-year TLB cycle will be highly challenging. Since my training set only contains 53 annual observations—meaning this 12-year cycle only repeats about four times in the historical data—there is a severe risk of overfitting. If we try to expand our basic models to capture such a long-term periodic pattern, the model might simply memorize the historical noise. Therefore, I will need to be extremely careful when evaluating the final model complexity to ensure it actually predicts the future rather than just repeating the past.

A Statistical Appendix

A.1 Autocorrelation and Partial Autocorrelation Functions

The Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) are fundamental statistical tools utilized in the EDA phase for ARIMA model identification.

Autocorrelation Function (ACF):
The ACF measures the linear dependence between an observation at time t and another observation at lag k. For a stationary time series x_t, the autocovariance at lag k is defined as $\gamma_k = \text{Cov}(x_t, x_{t-k})$. The ACF, denoted as ρ_k, is the normalized autocovariance:

$$\rho_k = \frac{\gamma_k}{\gamma_0}$$

where γ_0 represents the variance of the time series.

Partial Autocorrelation Function (PACF):
The PACF, denoted as ϕ_{kk}, isolates the direct correlation between x_t and x_{t-k} by removing the linear effects of the intervening observations $(x_{t-1}, x_{t-2}, \ldots, x_{t-k+1})$. It is formally defined as the correlation of the residuals obtained after regressing x_t and x_{t-k} on the intermediate lags:

$$\phi_{kk} = \text{Corr}(x_t - \hat{x}_t, x_{t-k} - \hat{x}_{t-k})$$

where $\hat{x}_t$ and $\hat{x}_{t-k}$ are the linear projections onto the space spanned by the intervening variables.

A.2 ARIMA Model Formulation

The preliminary model proposed for TFR is an ARIMA(1, 1, 1) model. Let x_t denote the observed value at time t. Using the backward shift operator B (where $Bx_t = x_{t-1}$), the equation expands to:

$$(1 - \phi_1 B)(1 - B)x_t = (1 + \theta_1 B)w_t$$

Where:

- $(1 - B)x_t = x_t - x_{t-1}$ represents the first-order differencing.
- ϕ_1 is the autoregressive (AR) parameter of order 1.
- θ_1 is the moving average (MA) parameter of order 1.
- $w_t \sim WN(0, \sigma_w^2)$ is the white noise error term.

A.3 Ljung-Box Portmanteau Test

The Ljung-Box test is utilised to rigorously check the adequacy of the fitted time series models by examining the overall randomness of the residuals. The hypotheses are defined as follows:

- H_0**:** The residuals are independently distributed (i.e., they resemble white noise, indicating a well-fitted model).

- H_1**:** The residuals are not independently distributed (i.e., they exhibit serial correlation, indicating an inadequate model).

A test statistic is calculated, and a p-value > 0.05 implies a failure to reject the null hypothesis, formally confirming that the model residuals are statistically white noise.